4 A rigid body is acted upon by a number of coplanar forces. Which of the following statements is correct?

A When the vector sum of all the forces acting on the body is zero, the object must be in a state of static equilibrium.

B When the coplanar forces are a couple, the magnitude of the torque of the couple on the object does not depend on the choice of the pivot point.

C When acted upon by three non-parallel coplanar forces whose lines of action intersect at a single point, the body must be in static equilibrium.

D When acted upon by two forces of equal magnitude and opposite directions, the body will always experience a net turning effect.

Ans: **B**

A couple creates a pure turning effect. Because the two constituent forces are equal in magnitude and opposite in direction, the distance variables to any arbitrary origin cancel out algebraically when you sum their moments. This makes the torque invariant across any reference point in space.

5 A ball is held above the ground and released. It falls to the ground and bounces several times